



HOW TO USE THIS SET

20 questions, mixed difficulty, 11 case-based. Attempt every question before checking the answer key on the last page.

● 4 Easy

● 10 Medium

● 6 Hard

1.

EASY

Which region of the small intestine is the shortest?

- A. Jejunum (2.5 m)
- B. All three regions are equal in length
- C. The ileocecal segment
- D. Duodenum (25 cm)
- E. Ileum (3.5 m)

2.

EASY

What controls the entry of material from the ileum into the large intestine?

- A. The plicae circulares
- B. The internal anal sphincter
- C. The muscularis mucosae
- D. The ileocecal valve, a sphincter
- E. The pyloric sphincter

3.

EASY

The brush border of an intestinal absorptive cell consists of:

- A. villi
- B. microvilli
- C. cilia
- D. secretory granules
- E. plicae circulares

4.

EASY

Which crypt cells secrete lysozyme?

- A. Paneth cells
- B. Enteroendocrine cells
- C. Absorptive cells
- D. Goblet cells
- E. M cells

5.

MEDIUM

CASE

A sample shows leaf-shaped villi, few goblet cells, and glands lying in the submucosa beneath the muscularis mucosae.

Which segment is this?

- A. Duodenum
- B. Pyloric stomach
- C. Ascending colon
- D. Jejunum
- E. Ileum

6.

MEDIUM

Plicae circulares are folds made of:

- A. mucosa and submucosa
- B. epithelium only
- C. all four tunics
- D. mucosa only
- E. muscularis externa and serosa

7.

MEDIUM CASE

At the base of an intestinal crypt lies an exocrine cell with basophilic cytoplasm and eosinophilic granules in its apical cytoplasm.

Which cell is this?

- A. Goblet cell
- B. Enteroendocrine cell
- C. Absorptive cell
- D. M cell
- E. Paneth cell

8.

MEDIUM

The connective-tissue core of a villus contains:

- A. mesothelium on loose CT
- B. smooth muscle bundles only
- C. dense CT with Meissner's plexus and glands
- D. Peyer's patch lymphoid nodules with germinal centres
- E. reticular tissue with capillaries and a lymphatic vessel

9.

MEDIUM CASE

An epithelial cell overlying lymphoid follicles in the ileum has a discontinuous basement membrane and endocytoses antigens, passing them to underlying macrophages and lymphocytes.

Which cell is this?

- A. Paneth cell
- B. Stem cell
- C. Enteroendocrine cell
- D. Goblet cell
- E. M cell

10.

MEDIUM

How does the number of goblet cells change along the small intestine?

- A. Goblet cells are absent from the ileum
- B. It stays the same in all regions
- C. It is highest in the jejunum
- D. It increases from the duodenum toward the ileum
- E. It decreases from the duodenum toward the ileum

11.

MEDIUM CASE

A triangular cell in the lower third of a crypt contains granules; its secretions regulate intestinal motility, peristalsis, secretions, visceral sensation and appetite.

Which set of products does this cell release?

- A. Pepsinogen and gastric lipase
- B. Lysozyme, phospholipase A2 and defensins
- C. Serotonin, somatostatin, CCK and secretin
- D. Alkaline mucus with bicarbonate
- E. Mucin and mucus for lubrication

12.

MEDIUM

Where are the epithelial stem cells of the small intestine, and how quickly does the epithelium turn over (per this lecture)?

- A. Base of the crypt; about 4–7 days
- B. Base of the crypt; about 2–4 days
- C. Tip of the villus; about 4–7 days
- D. Submucosa; about 3–4 days
- E. Tip of the villus; about 2–4 days

13.

MEDIUM

Compared with the duodenum, the jejunum has:

- A. club-shaped villi and Peyer's patches
- B. no villi and deep crypts
- C. long finger-shaped villi and many plicae circulares
- D. leaf-shaped villi and Brunner's glands
- E. abundant goblet cells and Peyer's patches

14.

MEDIUM

CASE

A tall columnar cell lining a villus has an oval nucleus at its base and a brush border at its apex.

Which statement best describes this cell?

- A. Absorption; ~3,000 microvilli enlarge its surface
- B. Release of hormones regulating appetite
- C. Endocytosis of antigens over lymphoid follicles
- D. Secretion of lysozyme into the crypt lumen
- E. Mitotic renewal of the epithelium

15.

HARD

CASE

A genetic defect abolishes the function of the cells at the base of the intestinal crypts that carry eosinophilic apical granules.

What is the most likely consequence?

- A. Reduced release of hormones regulating appetite
- B. Lack of alkaline mucus in the duodenum
- C. Failure to absorb nutrients via the brush border
- D. Loss of antigen sampling over Peyer's patches
- E. Less antimicrobial defence and crypt regulation

16.

HARD

CASE

Three sections: A shows club-shaped villi with abundant goblet cells and lymphoid nodules. B shows long finger-shaped villi with many plicae circulares. C shows leaf-shaped villi over submucosal glands.

Which order runs from proximal to distal?

- A. B, C, A
- B. A, C, B
- C. A, B, C
- D. C, B, A
- E. B, A, C

17.

HARD

CASE

Bacterial antigens in the ileal lumen contact the epithelium over a Peyer's patch.

Which sequence describes the fate of a luminal antigen at this site?

- A. Destroyed by enteroendocrine cell secretions
- B. Taken up by Paneth cells and stored in granules
- C. Endocytosed by M cells, then passed to macrophages and lymphocytes
- D. Absorbed by goblet cells and secreted in mucus
- E. Absorbed by absorptive cells and carried only by capillaries

18.

HARD

CASE

In a patient the duodenal submucosal glands secrete much less than normal.

What would be expected?

- A. Loss of the plicae circulares in the jejunum
- B. More acid on the duodenal lining; poorer pH for pancreatic enzymes
- C. Fewer brush-border microvilli on absorptive cells
- D. Bile and pancreatic ducts fail to open into the duodenum
- E. Abnormal antigen uptake by M cells in the ileum

19.

HARD

CASE

Disease flattens the plicae circulares of the jejunum.

What is the expected result?

- A. Loss of goblet cells from the mucosa
- B. Loss of Peyer's patches from the ileum
- C. Faster transit and less contact time for absorption
- D. No change, since villi alone control transit
- E. Slower transit and more contact time

20.

HARD

CASE

In one crypt: cell A lies at the base with eosinophilic apical granules; cell B is triangular and lies in the lower third; cell C is dividing at the base and gives rise to the other epithelial cells.

Which identification is correct?

- A. A = M cells, B = enteroendocrine cells, C = stem cells
- B. A = Paneth cells, B = enteroendocrine cells, C = stem cells
- C. A = stem cells, B = Paneth cells, C = enteroendocrine cells
- D. A = Paneth cells, B = stem cells, C = enteroendocrine cells
- E. A = enteroendocrine cells, B = Paneth cells, C = stem cells

ANSWER KEY & RATIONALE

Don't peek until you've committed to an answer on every question.

Q	Ans	Why
1	D	The duodenum is about 25 cm; the jejunum 2.5 m and the ileum 3.5 m.
2	D	The small intestine ends at the ileocecal valve, a sphincter controlling entry into the large intestine.
3	B	The brush border is the microvilli (~3,000 per absorptive cell) — surface cytoplasmic extensions.
4	A	Paneth cells (crypt base) contain eosinophilic granules with lysozyme.
5	A	Leaf-shaped villi, few goblet cells and submucosal (Brunner's) glands identify the duodenum.
6	A	Plicae circulares consist of mucosa + submucosa and slow the passage of contents, aiding absorption.
7	E	Basal crypt location, basophilic cytoplasm and eosinophilic apical lysozyme granules = Paneth cell.
8	E	The villus core is reticular tissue containing numerous blood capillaries and a lymphatic vessel.
9	E	M cells overlie Peyer's patches, have a discontinuous basement membrane and endocytose antigens.
10	D	Few in the duodenum, intermediate in the jejunum, abundant in the ileum.
11	C	Enteroendocrine cells (lower crypt, triangular) secrete serotonin, somatostatin, CCK, peptides and secretin.
12	B	Stem cells sit at the crypt base and undergo mitosis; most GI epithelial cells turn over in 2–4 days.
13	C	Jejunum: long finger-shaped villi, many plicae circulares, intermediate goblet cells.
14	A	Tall columnar cells with a basal oval nucleus and a brush border are the absorptive cells.
15	E	Paneth-cell products (lysozyme, phospholipase A2, defensins) attack microbial membranes / cell walls and regulate the crypt microenvironment.
16	D	Duodenum (C) → jejunum (B) → ileum (A).
17	C	M cells endocytose antigens and transport them to the underlying macrophages / lymphocytes — a protective function.
18	B	Brunner's glands secrete alkaline mucus that protects the lining from chyme acidity and sets the pH for pancreatic enzymes.
19	C	Circular folds slow the passage of contents through the small intestine, thus facilitating absorption.
20	B	Apical eosinophilic granules (lysozyme) = Paneth; triangular hormone cells in the lower third = enteroendocrine; mitotic cells at the base = stem cells.

Histology GIT Block · MEDucation by Mattin Majid · Source: GIT Histology SGL 2, Dr Hero K. Mustafa
Written from the lecture slides only – check any answer you doubt against your lecture notes.